

# Roman Penunuri

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## SUMMARY

Fourth-year Electrical Engineering student with a strong foundation in analog circuit design, electronics, and bench characterization. Proficient iterating between simulation environments (Falstad, LTSpice) and physical hardware, with hands-on bench experience using oscilloscopes, function generators, multimeters, DC power supplies, and resistance decade boxes. Completed KiCad schematic capture and PCB layout through an IEEE design workshop. Industry experience developing CT wiring schematics and electrical documentation for industrial systems. Self-motivated and driven to see technical problems through from start to finish, proactive about seeking input, and comfortable integrating ideas across disciplines.

## EDUCATION

### University of Houston, Cullen College of Engineering

Houston, TX

B.S. Electrical Engineering, Power Systems Concentration | GPA: 3.82

May 2028

Dean's List: Fall 2024, Spring 2026

**Relevant Coursework:** Electronics (ECE 3355), Electronics Laboratory (ECE 3155), Circuit Analysis I & II, Circuit Analysis Laboratory, Digital Logic Design, Applied Electromagnetic Waves (Honors), Signals & Systems Analysis, Microprocessor Systems

## SKILLS

**Certifications:** OSHA-30 General Industry | Pursuing ETAP Operator Certification

**Proficient:** Falstad, LTSpice, KiCad (schematic capture and PCB layout), oscilloscopes, function generators, multimeters, DC power supplies, resistance decade boxes, MATLAB, C

**Currently Learning:** Python, ETAP, AutoCAD Electrical

**Languages:** Fluent in English, Conversational in Spanish

## PROJECTS

### Water Level Indicator: BJT Sensing Circuit

University of Houston

- Designed and bench-characterized a BJT transistor-based water level sensing circuit using oscilloscopes, function generators, and DC power supplies; diagnosed LED underperformance by iterating between Falstad simulation and physical node probing to trace the failure to transistor saturation caused by unexpectedly high distilled water resistivity, resolving the fault through systematic simulation-hardware iteration

### 555 Timer PCB: KiCad Design Workshop

IEEE UH Chapter

- Completed full KiCad design workflow for a 555 timer astable circuit, progressing from schematic capture and component placement through PCB layout and Gerber file generation, as part of an IEEE design workshop

## WORK EXPERIENCE

### Air Liquide

Pasadena, TX

Plant Operations Electrical Engineering Intern

June 2025 - August 2025

- Developed CT wiring schematics for industrial bus heater ammeter monitoring circuits and authored scope-of-work documentation specifying installation requirements and testing procedures for field technicians and outside contractors
- Migrated and validated motor protection relay settings (GE Vernova Multilin 469 to 869) in EnerVista, verifying protection element configurations to eliminate recurring equipment faults and reduce projected maintenance costs by ~\$10,000
- Reviewed plant electrical systems and one-line diagrams during Channelview turnaround, developing familiarity with industrial switchgear, motor circuits, and relay coordination in live production environments
- Selected for return internship (Summer 2026) based on performance and documented project impact

### Take 5 Oil Change

Crosby, TX

Certified Lube Technician

March 2025 - May 2025

- Executed 40+ vehicle service cycles daily to manufacturer specifications, maintaining component-level precision and procedural discipline in a fast-paced environment

### Planet Fitness

Crosby, TX

Member Services Representative

January 2024 - March 2025

- Managed operations for a 1,000+ member facility while converting an average of 6 prospects per week through structured presentations and relationship-driven communication

## ORGANIZATIONS

### Tau Beta Pi Honor Society

Houston, TX

Member

April 2025 - Present

### Society of Hispanic Professional Engineers (SHPE)

Houston, TX

Member

August 2023 - Present

### IEEE University of Houston Chapter

Houston, TX

Member

August 2025 - Present